Nandini Gulhane

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Summary

Final-year Information Technology student with a strong foundation in Java, object-oriented programming, and backend development. Experienced with MySQL, REST APIs. Focused on building secure, maintainable systems and continuously improving through practical, hands-on development projects.

Education

Vishwakarma Institute of Technology — Pune, Maharashtra
Information Technology Bachelor of Technology — 2022 - 2026
CGPA: 9.01

Skills

Programming Languages:	Java, Python, SQL, Spring Boot
Developer Tools:	Git, GitHub, Docker, AWS (EC2, IAM), Linux

Projects

Smart Home Automation using Facial Recognition *Java, MySQL, OpenCV, Python, MicroPython*

- Designed and created an intelligent home automation system using facial recognition, achieving a 95% accuracy rate.
- Integrated the system with 5+ IoT devices (smart lights, speakers, etc.), reducing manual efforts by 80%.
- Optimized MySQL queries, improving database efficiency by 40%.
- **Challenges Faced:** Ensured real-time facial recognition while maintaining system efficiency for smart home automation.

MediConnect – AI-powered Doctor–Patient Platform *Python, Flask, OCR, Pinecone, RAG, LangChain*

- Developed a secure healthcare platform enabling patients to upload medical documents via SSO authentication and share access with doctors through OTP-based authorization.
- Implemented OCR processing to extract text from uploaded reports, vectorized the data, and stored embeddings in Pinecone for efficient retrieval.
- Processed and vectorized 200+ medical documents using OCR with 92% text extraction accuracy for downstream RAG.
- Built a RAG-powered AI assistant that answers doctor queries using context retrieved through vector similarity search and cosine similarity scoring.
- Enabled doctor–patient interactions including AI-assisted report analysis, medical query answering, and automated appointment reminders.
- Designed a standardized medical summary generator that consolidates all reports into a structured, readable format for doctors.

Email Threat Detection System using LLMs *Python, Flask, LangChain, Pinecone, SQLAlchemy, VirusTotal API*

- Built a backend system to detect phishing and spam emails using large language models (LLMs) with over 90% accuracy.
- Integrated IMAP for email fetching, VirusTotal for URL/file reputation scanning, and Pinecone for vector-based semantic similarity search.
- Utilized LangChain for prompt chaining and retrieval-augmented generation (RAG) to classify threats with high accuracy.
- Implemented persistent UID tracking via SQLite to avoid duplicate processing across sessions.